EIN 4343 Inventory and Supply Chain Systems

Prof. Elif Akçalı

Logistics decisions

- Transportation mode selection
- Shipment size specification
- Carrier selection

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EIN4343 INVENTORY & SUPPLY CHAIN SYSTEMS

LOGISTICS: CONSOLIDATION

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Consolidation

Forms of consolidation in supply chain systems

- Inventory consolidation
- Warehouse consolidation
- Shipment consolidation

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Inventory consolidation

- Consolidation of inventory from multiple stock locations to a single stock location
- Pro: Reduced inventory holding cost for the same service level (under some mild assumptions)
- Con: Increased transportation cost

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A primer: Sums of random variables: Expected value function

For any two random variables X and Y, the expected value of the sum of those variables is equal to the sum of their expected values.

That is, we have

$$E[X + Y] = E[X] + E[Y]$$

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A primer: Sums of random variables: Variance function

For any two random variables X and Y, the variance of the sum of those variables is equal to the sum of their variances plus twice the covariance.

That is, we have

$$\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\mathrm{Cov}(X+Y)$$

<u>A special case</u>: If X and Y are <u>independent</u> random variables, then we have

$$\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$$

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Example 7.4.a (1 of 3)

- MedEq, located in Elmira, New York is a manufacturer of supply kits that are used by doctors in cosmetic procedures.
- The kits weigh 0.5 lbs and cost $200 each. MedEq currently has 24 sales territories (i.e., regions) throughout the U.S. such that the demand in each area is distributed identically and independently.
- Furthermore, each sales territory has its own sales force and a local distribution center where all MedEq supply kit inventory is stored.
- MedEq uses an annual inventory carrying rate of 40 percent at the local distribution centers and you may assume that there are 48 weeks in a year.

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Example 7.4.a (2 of 3)

- Currently, the 24 local distribution centers are replenished from Elmira every four weeks using Yellow Transportation. MedEq outsources the shipment of supply kits to Yellow Transportation.
- Yellow Transportation charges MedEq a transportation rate of $\$40 + 1.1x$, where x is the weight of the shipment in pounds.
- The average lead time from Elmira to any of the local distribution centers is about one week.
- Weekly demand for MedEq supply kits at each territory follows a normal distribution approximately with a mean of 40 units and a standard deviation of 10 units.

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Example 7.4.a (3 of 3)

- Weekly demand for MedEq supply kits at each territory follows a normal distribution approximately with a mean of 40 units and a standard deviation of 10 units.
- MedEq targets a Type I service level of 95% in each sales territory.
- What is the total annual logistics cost associated with the current operating policy? You may ignore in-transit inventory holding costs.

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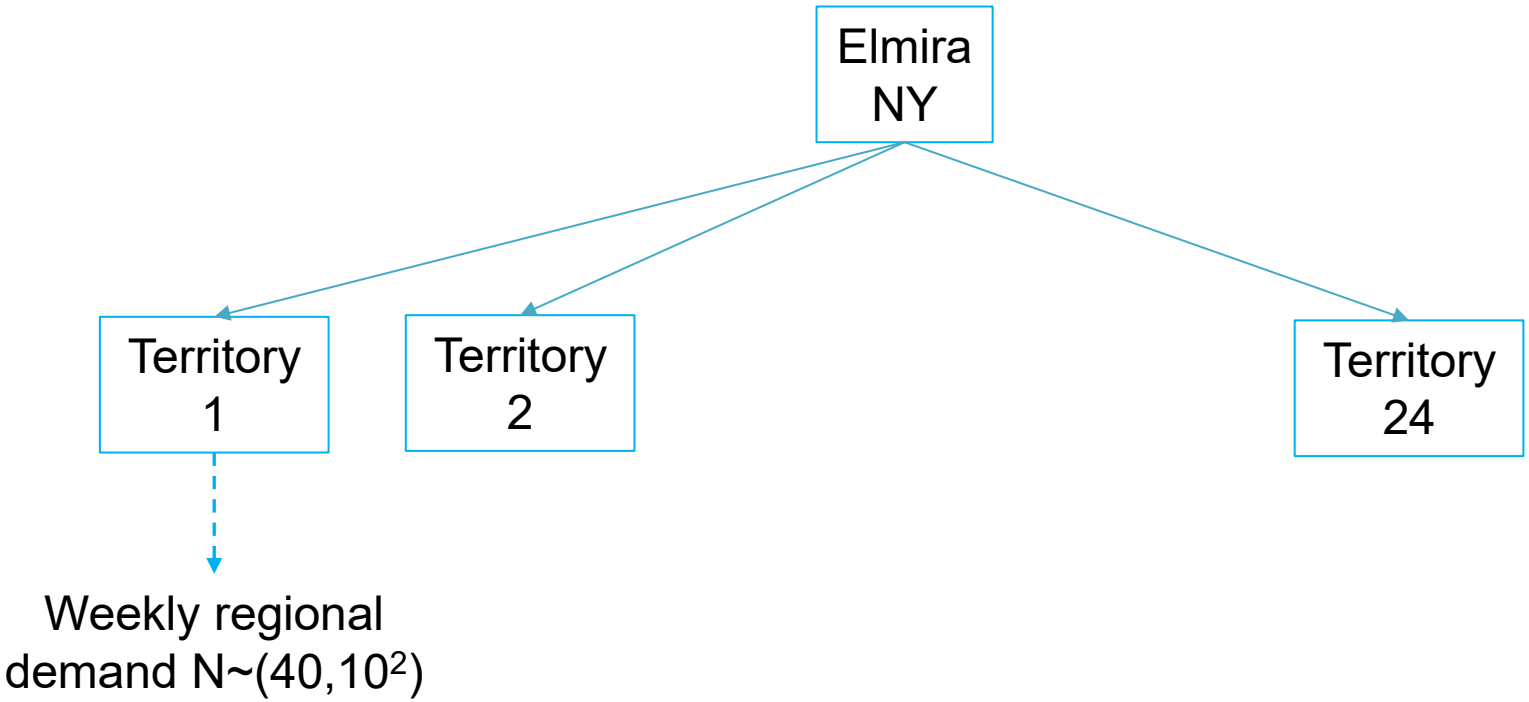
Example 7.4.a



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Example 7.4.a

Annual demand for region i: $\lambda_i = (48)(40) = 1{,}920 \text{ units/yr}$

Order quantity for region i: $q_i = (4)(40) = 160 \text{ units/order}$

Mean DDLT for region i: $\mu_{L,i} = \tau\mu_i = (1)(40) = 40 \text{ units}$

Std Dev DDLT for region i: $\sigma_{L,i} = \sqrt{\tau\sigma_i^2} = \sqrt{(1)(10^2)} = 10 \text{ units}$

Type I SL for region i: $\alpha_i = 0.95 \Longrightarrow z_i = 1.6449$

Reorder level for region i: $R_i = \mu_{L,i} + z_i\sigma_{L,i} = 40 + (1.6449)(10) = 56.45 \text{ units}$

Safety stock for region i: $s_i = R_i - \mu_{L,i} = 56.45 - 40 = 16.45 \text{ units}$

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Example 7.4.a

Annual Total Transportation Cost (ATTC)

$$ATTC(q_1, \dots, q_{24}) = (24)\left(\frac{1{,}920}{160}\right)\big(40 + (1.1)(160)(0.5)\big) = (24)(\$1{,}536/\text{year})$$

(# of sales territories) (# of shipments/year/territory) (Transportation cost per shipment)

Annual Onsite Inventory Holding Cost (AOIHC)

$$AOIHC(q_1, \dots, q_{24}) = (24)(0.40)(200)\left[16.45 + \frac{160}{2}\right] = (24)(\$7{,}716/\text{year})$$

(# of sales territories) (Inventory holding cost/unit/year) (Average inventory level)

Total Cost

$$ATTC(q_1, \dots, q_{24}) + AOIHC(q_1, \dots, q_{24}) = \$222{,}048/\text{year}$$

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Example 7.4.b

- MedEq considers eliminating the local distribution centers and consolidating all its inventories at the local distribution centers at a finished goods warehouse in Louisville, Kentucky.

- This central warehouse is to be replenished from the manufacturing facility each week, and customer orders are to be shipped directly from the central warehouse to the customers via FedEx.

- FedEx has made an offer to charge MedEq $\$17 + 2.50x$ per shipment, where x is the quantity shipped in pounds.

- The average customer order size is 8 units.

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Example 7.4.b

- The lead time to replenish this central warehouse from MedEq's manufacturing facility would be one week, and each replenishment trip would cost the company $500.
- MedEq would still target a Type I service level of 95 percent.
- Due to the incentives given to medical businesses in Kentucky, MedEq will incur an annual inventory carrying charge of 15 percent per year at this warehouse.
- What is the total annual logistics cost associated with this proposed operating policy?

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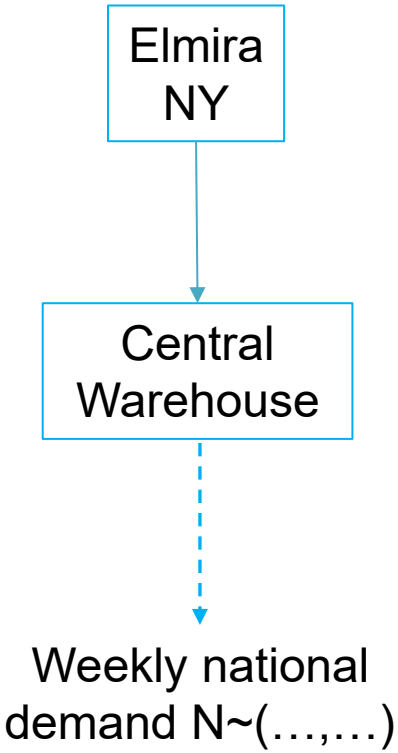
Example 7.4.b



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Reminder: Sum of iid random variables

- Consider n iid random variables X_i for $\mathrm{i} = 1, \ldots, n$ each with expected value μ and variance σ^2.

- Suppose a random variable, X_S, that is formed by the summation of these n iid random variables. Then, we have the following:

$$E[X_S] = E[X_1 + X_2 + \cdots + X_n]$$
$$= E[X_1] + E[X_2] + \ldots E[X_n] = n\mu$$

$$\mathrm{Var}(X_S) = \mathrm{Var}(X_1 + X_2 + \cdots + X_n)$$
$$= \mathrm{Var}(X_1) + \mathrm{Var}(X_2) + \cdots + \mathrm{Var}(X_n) = n\sigma^2$$

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Example 7.4.b

Annual demand: $\lambda = (24)(48)(40) = 40{,}080 \text{ units/year}$

Order quantity for the warehouse: $q_w = (1)(24)(40) = 960 \text{ units/replenishment}$

Mean weekly demand for the warehouse: $\mu_w = (24)(40) = 960 \text{ units}$

Std Dev weekly demand for the warehouse: $\sigma_w = \sqrt{(24)\sigma_i^2} = \sqrt{(24)10^2} = 48.99 \text{ units}$

Mean DDLT for the warehouse: $\mu_{L,w} = \tau\mu_w = (1)(960) = 960 \text{ units}$

Std Dev DDLT for the warehouse: $\sigma_{L,w} = \sqrt{\tau\sigma_w^2} = \sqrt{(1)48.99^2} = 48.99 \text{ units}$

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Example 7.4.b

Type I SL for the warehouse: $\alpha_w = 0.95 \Longrightarrow z_w = 1.6449$

Reorder level for the warehouse: $R_w = \mu_{L,w} + z_w \sigma_{L,w}$

$$= 960 + (1.6449)(48.99)$$

$$= 1{,}040.58 \text{ units}$$

Safety stock for the warehouse: $s_w = R_w - \mu_{L,w}$

$$= 1040.58 - 960$$

$$= 80.58 \text{ units}$$

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Example 7.4.b

b) What is the total annual logistics costs associated with the proposed operating policy?

Annual Total Transportation Cost (ATTC)

$$ATTC(q_w) = 500\frac{46{,}080}{960} + [17 + (2.5)(8)(0.5)]\frac{46{,}080}{8} = \$24{,}000 + \$155{,}520/\text{year}$$

Annual Onsite Inventory Holding Cost (AOIHC)

$$AOIHC(q_w) = (0.15)(200)\left[80.58 + \frac{960}{2}\right] = \$16{,}817/\text{year}$$

Total Cost

$$ATTC(q_w) + AOIHC(q_w) = \$196{,}337/\text{year}$$

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MedEq: Comparison

	Via 24 regional distribution centers	Via a national warehouse
Transportation	$36,864	$175,520
On-Site Inv.	$185,184	$16,817
Total	$222,048	$196,337

	Via 24 regional distribution centers	Via a national warehouse
Safety stock level	394.8 units	80.58 units

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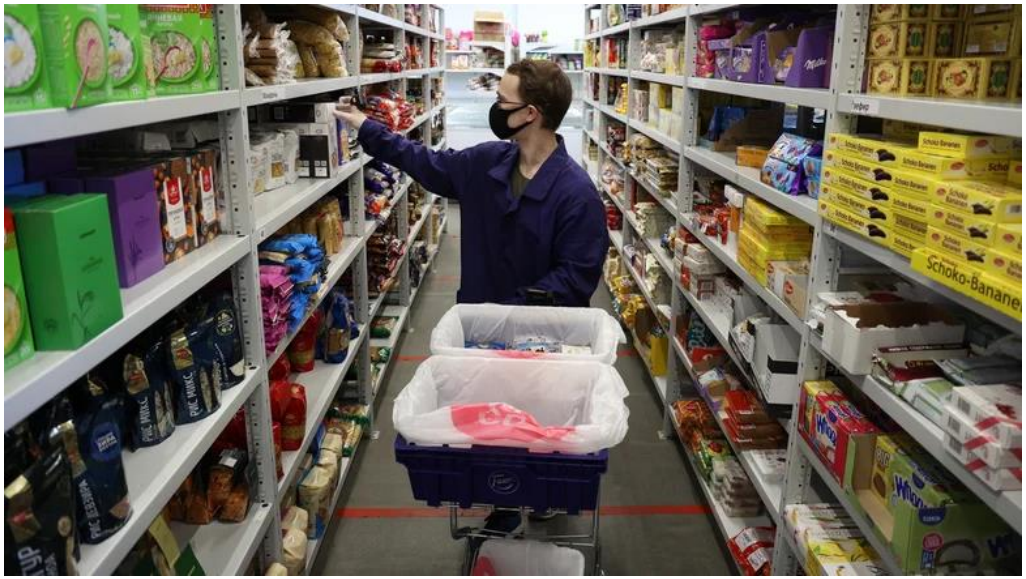
Dark stores: Micro fulfillment centers



“In early 2022, there were 115 dark stores in New York City's five boroughs, per a study by BetaNYC. Gorillas, a prominent dark store delivery chain, has expanded to New York, Boston, and Chicago, and high-traffic competitors like Gopuff have emerged alongside.”

Read More: https://www.tastingtable.com/1515571/dark-grocery-store-explained/

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Gopuff @ GNV

Not really a “dark” store…

Consolidation

Forms of consolidation in supply chain systems

- Inventory consolidation
- Warehouse consolidation
- Shipment consolidation

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Warehouse consolidation

- Consolidation of incoming shipments from multiple suppliers to a stock location prior to distribution to customers
- Pro: Demand consolidation; shorter haul distances; smaller, customized loads to customers; exploit cross-docking opportunities
- Con: Increased control complexity; additional layer of inventory (if not fully cross-docked)

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Example 7.5 (1 of 2)

- Solestica, a manufacturer of integrated circuits and systems, operates two assembly plants in Los Angeles and San Jose, CA.
- These plants operate for 360 days per year, and each produces cell phone boards at a rate of 100 units per day.
- Each board requires one unit of two major components, a backboard and an integrated circuit kit that includes all the components needed.
- The boards are produced at a plant in Chicago, IL and the component kits are assembled at a plant in St. Petersburg, FL.

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Example 7.5 (2 of 2)

- The boards and the kits are shipped in full truckloads from the manufacturing plants to assembly plants in California directly.
- The travel times for each truck is 6 days, and the capacity of a truck is 4,000 units (regardless of the type of the component shipped).
- Solestica pays $3,500 per full truckload.
- Inventory holding costs are incurred for in-transit and on-site at the assembly plants, and the inventory holding cost is $0.40/unit/day.

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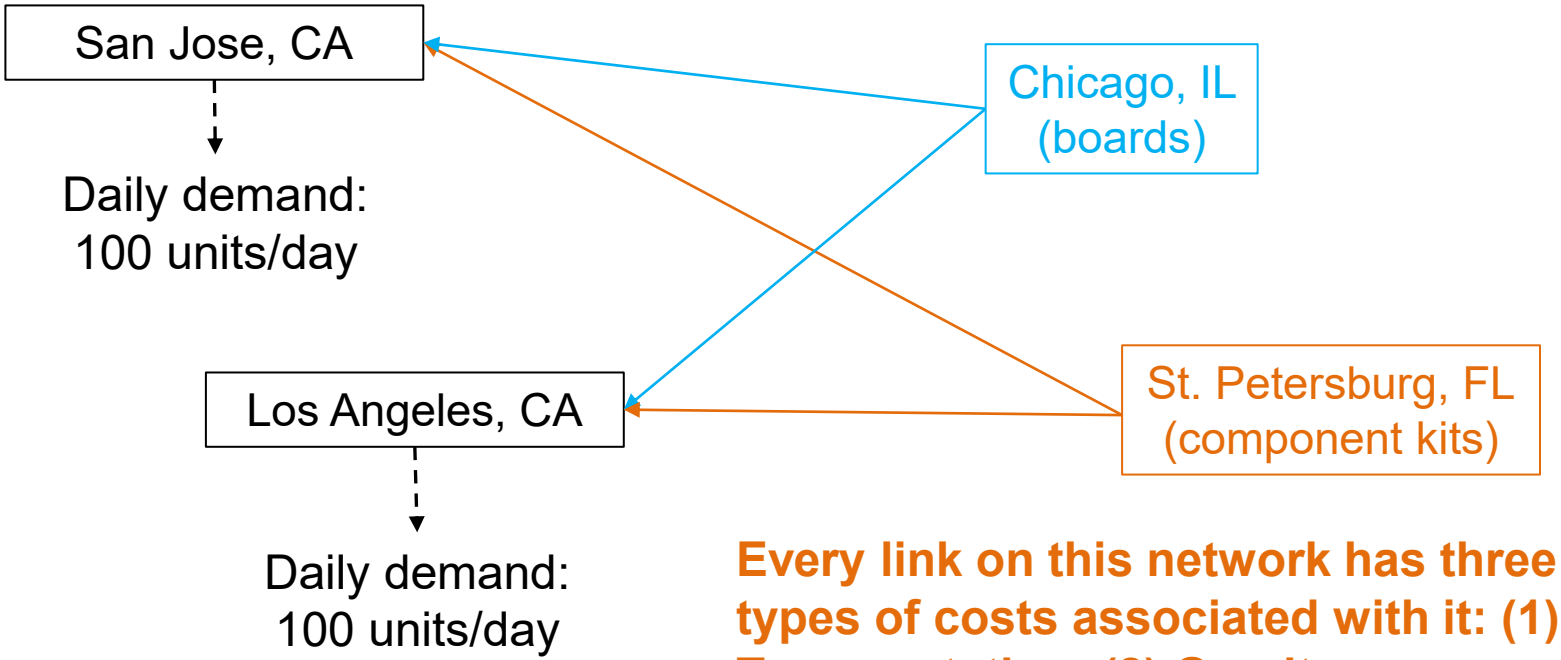
Example 7.5.a

What is the annual total logistics cost (including transportation, in-transit and on-site inventory costs) associated with the current policy of shipping directly in full truckloads?



Every link on this network has three types of costs associated with it: (1) Transportation; (2) On-site inventory and (3) In-transit inventory.

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Example 7.5.a

Annual Total Transportation Cost (ATTC)

$$ATTC_{StP\rightarrow SJ} = \frac{K\lambda_{StP\rightarrow SJ}}{q_{StP\rightarrow SJ}} = \frac{(3{,}500)(360)(100)}{4{,}000} = \$31{,}500/\text{year}$$

$$ATTC_{StP\rightarrow LA} = \frac{K\lambda_{StP\rightarrow LA}}{q_{StP\rightarrow LA}} = \frac{(3{,}500)(360)(100)}{4{,}000} = \$31{,}500/\text{year}$$

$$ATTC_{Chi\rightarrow SJ} = \frac{K\lambda_{Chi\rightarrow SJ}}{q_{Chi\rightarrow SJ}} = \frac{(3{,}500)(360)(100)}{4{,}000} = \$31{,}500/\text{year}$$

$$ATTC_{Chi\rightarrow LA} = \frac{K\lambda_{Chi\rightarrow LA}}{q_{Chi\rightarrow LA}} = \frac{(3{,}500)(360)(100)}{4{,}000} = \$31{,}500/\text{year}$$

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Example 7.5.a

Annual Onsite Inventory Holding Cost (AOIHC)

$$AOIHC_{kits\ @\ SJ} = \frac{hq_{StP \to SJ}}{2} = \frac{(0.4)(360)(4{,}000)}{2} = \$288{,}000/year$$

$$AOIHC_{brds\ @\ SJ} = \frac{hq_{Chi \to SJ}}{2} = \frac{(0.4)(360)(4{,}000)}{2} = \$288{,}000/year$$

$$AOIHC_{kits\ @\ LA} = \frac{hq_{StP \to LA}}{2} = \frac{(0.4)(360)(4{,}000)}{2} = \$288{,}000/year$$

$$AOIHC_{brds\ @\ LA} = \frac{hq_{Chi \to LA}}{2} = \frac{(0.4)(360)(4{,}000)}{2} = \$288{,}000/year$$

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Example 7.5.a

Annual In−transit Inventory Holding Cost (AIIHC)

$$AIIHC_{kits\ @\ SJ} = h\lambda_{StP\to SJ}\tau_{StP\to SJ} = (0.4)(360)(100)(360)\left(\frac{6}{360}\right) = \$86{,}400/year$$

$$AIIHC_{brds\ @\ SJ} = h\lambda_{StP\to SJ}\tau_{StP\to SJ} = (0.4)(360)(100)(360)\left(\frac{6}{360}\right) = \$86{,}400/year$$

$$AIIHC_{kits\ @\ LA} = h\lambda_{StP\to SJ}\tau_{StP\to SJ} = (0.4)(360)(100)(360)\left(\frac{6}{360}\right) = \$86{,}400/year$$

$$AIIHC_{brds\ @\ LA} = h\lambda_{StP\to SJ}\tau_{StP\to SJ} = (0.4)(360)(100)(360)\left(\frac{6}{360}\right) = \$86{,}400/year$$

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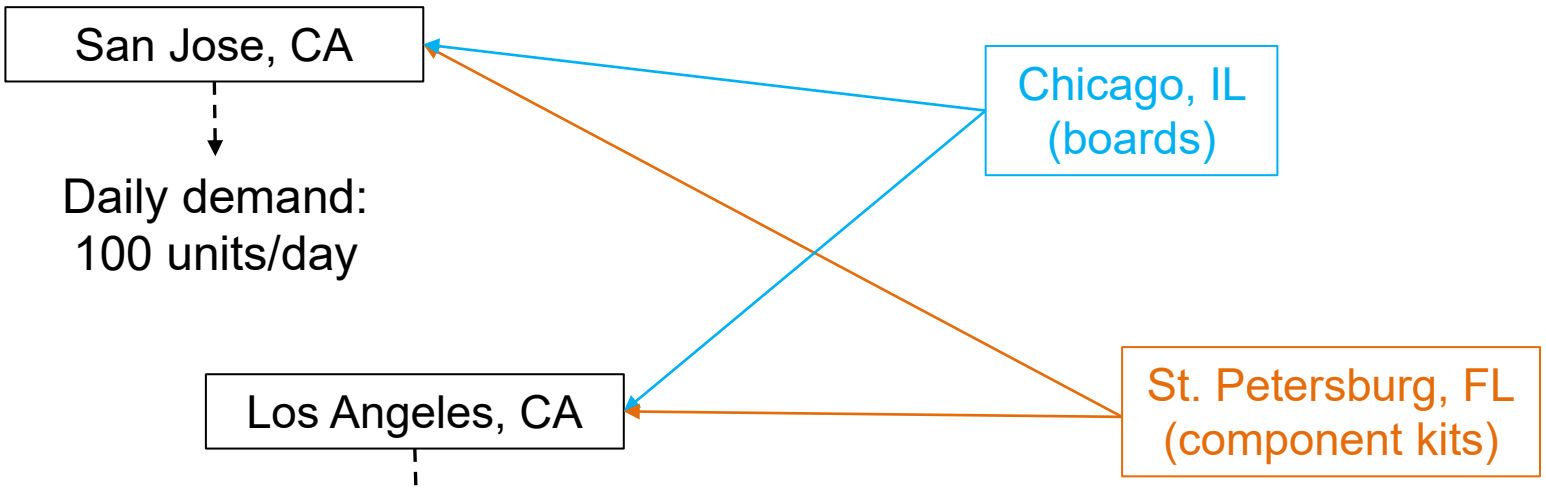
Example 7.5.b

What would the annual total logistics cost be, if Solestica were to use optimal shipment sizes from the manufacturing plants to the assembly plants?



Every link on this network has three types of costs associated with it: (1) Transportation; (2) On-site inventory and (3) In-transit inventory.

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Example 7.5.b

Optimal Shipment Sizes

$$q_{StP\to SJ} = \sqrt{\frac{2K\lambda_{StP\to SJ}}{h_{StP\to SJ}}} = \sqrt{\frac{2(3{,}500)(100)(360)}{(0.4)(360)}} = 1{,}322.88 \text{ units}$$

$$q_{StP\to LA} = \sqrt{\frac{2K\lambda_{StP\to LA}}{h_{StP\to LA}}} = \sqrt{\frac{2(3{,}500)(100)(360)}{(0.4)(360)}} = 1{,}322.88 \text{ units}$$

$$q_{Chi\to SJ} = \sqrt{\frac{2K\lambda_{Chi\to SJ}}{h_{Chi\to SJ}}} = \sqrt{\frac{2(3{,}500)(100)(360)}{(0.4)(360)}} = 1{,}322.88 \text{ units}$$

$$q_{Chi\to LA} = \sqrt{\frac{2K\lambda_{Chi\to LA}}{h_{Chi\to LA}}} = \sqrt{\frac{2(3{,}500)(100)(360)}{(0.4)(360)}} = 1{,}322.88 \text{ units}$$

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Example 7.5.b

Annual Total Transportation Cost (ATTC)

$$ATTC_{StP\to SJ} = \frac{K\lambda_{StP\to SJ}}{q_{StP\to SJ}} = \frac{(3{,}500)(360)(100)}{1{,}322.88} = \$95{,}247/\text{year}$$

$$ATTC_{StP\to LA} = \frac{K\lambda_{StP\to LA}}{q_{StP\to LA}} = \frac{(3{,}500)(360)(100)}{1{,}322.88} = \$95{,}247/\text{year}$$

$$ATTC_{Chi\to SJ} = \frac{K\lambda_{Chi\to SJ}}{q_{Chi\to SJ}} = \frac{(3{,}500)(360)(100)}{1{,}322.88} = \$95{,}247/\text{year}$$

$$ATTC_{Chi\to LA} = \frac{K\lambda_{Chi\to LA}}{q_{Chi\to LA}} = \frac{(3{,}500)(360)(100)}{1{,}322.88} = \$95{,}247/\text{year}$$

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Example 7.5.b

Annual Onsite Inventory Holding Cost (AOIHC)

$$AOIHC_{kits\ @\ SJ} = \frac{hq_{StP \to SJ}}{2} = \frac{(0.4)(360)(1{,}322.88)}{2} = \$95{,}247/\text{year}$$

$$AOIHC_{brds\ @\ SJ} = \frac{hq_{Chi \to SJ}}{2} = \frac{(0.4)(360)(1{,}322.88)}{2} = \$95{,}247/\text{year}$$

$$AOIHC_{kits\ @\ LA} = \frac{hq_{Chi \to LA}}{2} = \frac{(0.4)(360)(1{,}322.88)}{2} = \$95{,}247/\text{year}$$

$$AOIHC_{brds\ @\ LA} = \frac{hq_{StP \to LA}}{2} = \frac{(0.4)(360)(1{,}322.88)}{2} = \$95{,}247/\text{year}$$

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Example 7.5.b

Annual In−transit Inventory Holding Cost (AIIHC)

$$AIIHC_{kits\ @\ SJ} = h\lambda_{StP\to SJ}\tau_{StP\to SJ} = (0.4)(360)(100)(360)\left(\frac{6}{360}\right) = \$86{,}400/\text{year}$$

$$AIIHC_{brds\ @\ SJ} = h\lambda_{StP\to SJ}\tau_{StP\to SJ} = (0.4)(360)(100)(360)\left(\frac{6}{360}\right) = \$86{,}400/\text{year}$$

$$AIIHC_{kits\ @\ LA} = h\lambda_{StP\to SJ}\tau_{StP\to SJ} = (0.4)(360)(100)(360)\left(\frac{6}{360}\right) = \$86{,}400/\text{year}$$

$$AIIHC_{brds\ @\ LA} = h\lambda_{StP\to SJ}\tau_{StP\to SJ} = (0.4)(360)(100)(360)\left(\frac{6}{360}\right) = \$86{,}400/\text{year}$$

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Example 7.5.c

A consultant suggested Solestica use a warehouse in Topeka, KA to consolidate shipments for assembly plants and perform cross-docking using trucks with a smaller capacity.

Then, the boards and component kits would be shipped from the manufacturing plants to the warehouse in full truckloads, where the travel time for each truck is 2 days, truck capacity is 2,000 units and each truckload costs $1,000 per shipment.

Once every 10 days, a separate truck goes to each of the assembly plants in San Jose and Los Angeles from the warehouse in Topeka, carrying 1,000 boards and 1,000 component kits.

The travel time for this leg is 3 days and truck capacity is also 2,000 units with a transportation cost of $1,000 per truckload.

What would the annual total logistics cost associated with this policy be, if Solestica were to implement this network configuration and shipment policy?

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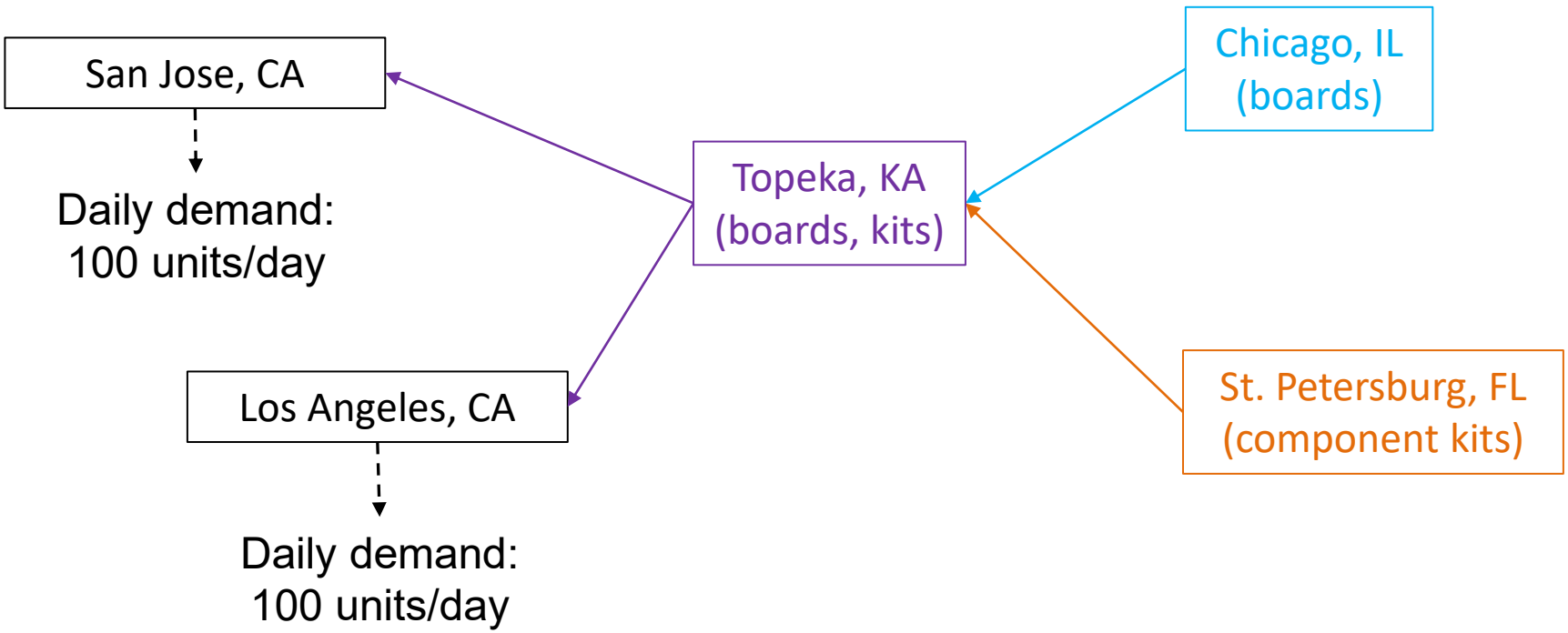
Example 7.5.c

Every link on this network has three types of costs associated with it: (1) Transportation; (2) On-site inventory and (3) In-transit inventory.



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Example 7.5.c

Annual Total Transportation Cost (ATTC)

$$ATTC_{brds\to W} = \frac{K\lambda_{brds\to SJ,LA}}{q_{brds\to SJ,LA}} = \frac{(1{,}000)(100+100)(360)}{2{,}000} = \$36{,}000/\text{year}$$

$$ATTC_{kits\to W} = \frac{K\lambda_{kits\to SJ,LA}}{q_{kits\to SJ,LA}} = \frac{(1{,}000)(100+100)(360)}{2{,}000} = \$36{,}000/\text{year}$$

$$ATTC_{W\to SJ} = \frac{K\lambda_{brds,kits\to SJ}}{q_{brds,kits\to SJ}} = \frac{(1{,}000)(100+100)(360)}{2{,}000} = \$36{,}000/\text{year}$$

$$ATTC_{W\to LA} = \frac{K\lambda_{brds,kits\to LA}}{q_{brds,kits\to LA}} = \frac{(1{,}000)(100+100)(360)}{2{,}000} = \$36{,}000/\text{year}$$

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Example 7.5.c

Annual Onsite Inventory Holding Cost (AOIHC)

$$AOIHC_{kits\ @\ SJ} = \frac{hq_{kits \to SJ}}{2} = \frac{(0.4)(360)(1{,}000)}{2} = \$72{,}000/\text{year}$$

$$AOIHC_{brds\ @\ SJ} = \frac{hq_{brds \to SJ}}{2} = \frac{(0.4)(360)(1{,}000)}{2} = \$72{,}000/\text{year}$$

$$AOIHC_{brds\ @\ LA} = \frac{hq_{brds \to LA}}{2} = \frac{(0.4)(360)(1{,}000)}{2} = \$72{,}000/\text{year}$$

$$AOIHC_{kits\ @\ LA} = \frac{hq_{kits \to LA}}{2} = \frac{(0.4)(360)(1{,}000)}{2} = \$72{,}000/\text{year}$$

$$AOIHC_{kits,brds\ @\ W} = 0$$

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Example 7.5.c

Annual In−transit Inventory Holding Cost (AIIHC)

$$AIIHC_{kits\ @\ SJ} = h\lambda_{StP\to SJ}\tau_{StP\to W\to SJ} = (0.4)(360)(100)(360)\left(\frac{3+2}{360}\right) = \$72{,}000/\text{year}$$

$$AIIHC_{brds\ @\ SJ} = h\lambda_{Chi\to SJ}\tau_{Chi\to W\to SJ} = (0.4)(360)(100)(360)\left(\frac{3+2}{360}\right) = \$72{,}000/\text{year}$$

$$AIIHC_{brds\ @\ LA} = h\lambda_{StP\to LA}\tau_{StP\to W\to LA} = (0.4)(360)(100)(360)\left(\frac{3+2}{360}\right) = \$72{,}000/\text{year}$$

$$AIIHC_{kits\ @\ LA} = h\lambda_{StP\to LA}\tau_{Chi\to W\to LA} = (0.4)(360)(100)(360)\left(\frac{3+2}{360}\right) = \$72{,}000/\text{year}$$

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Solestica: Comparison

	Direct shipment FTL	Direct shipment Optimal/LTL	Via warehouse FTL
Transportation	$126,000	$380,988	$144,000
On-Site Inv.	$1,152,000	$380,990	$288,000
In-Transit Inv.	$345,600	$345,600	$288,000
Total	$1,623,600	$1,107,578	$720,000

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Consolidation

Forms of consolidation in supply chain systems

- Inventory consolidation
- Warehouse consolidation
- Shipment consolidation

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Shipment consolidation

- Consolidation of
  - incoming shipments from multiple suppliers or
  - outgoing shipments to multiple customers
- Pro: Demand/supply consolidation; smaller, customized pick-up or delivery loads
- Con: Increased control complexity; increased transportation cost due to more frequent deliveries

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Example 7.7 (1 of 2)

- Whole Foods Market (WFM) is a retailer of natural and organic foods with more than stores in North America, Canada, and United Kingdom.
- The company operates several regional distribution centers in North America. The South Regional Office, located in Roswell, GA, is responsible for making replenishment decision for the stores located in Georgia and the Carolinas. There are six stores in Georgia, five in North Carolina and one in South Carolina that are served by this regional office.
- Dry goods (including sugar, flour, cereal and coffee among others) are replenished from the regional distribution center in batches. WFM operates three stores in Atlanta.

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Example 7.7 (2 of 2)

- Currently, the dry goods are shipped directly from the distribution center to the individual stores in full truckloads of capacity 2,000 lbs. WFM pays $2,250/truckload.

- The weekly demand for dry goods in these three stores are as

Store	Briarcliff (Brc)	Sandy Springs (SaS)	Ponce de Leon (PdL)
Average demand (lbs/week)	250	500	1,000

- WFM incurs a weekly inventory carrying and handling cost of $2.5/lb of dry goods. The stores operate 48 weeks/year. Since the distribution center is in Georgia, in-transit inventory holding costs incurred can assumed to be negligible.

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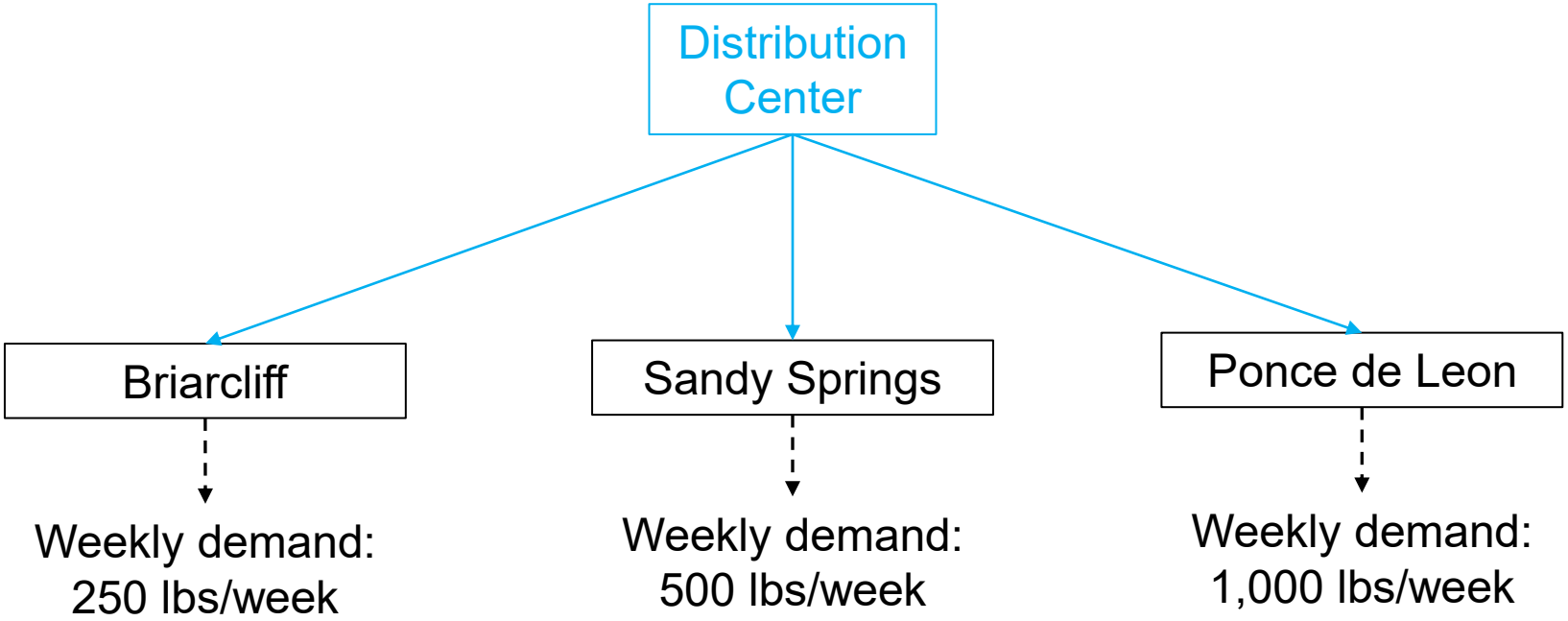
Example 7.7.a

What is the annual total logistics cost (including transportation and on-site inventory costs) associated with the current policy?



Every link on this network has two types of costs associated with it: (1) Transportation; and (2) On-site inventory.

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Example 7.7.a

Annual Total Transportation Cost (ATTC)

$$ATTC_{DC\rightarrow Brc} = \frac{K\lambda_{Brc}}{q_{DC\rightarrow Brc}} = \frac{(2{,}250)(250)(48)}{2{,}000} = \$13{,}500/\text{year}$$

$$ATTC_{DC\rightarrow SaS} = \frac{K\lambda_{SaS}}{q_{DC\rightarrow SaS}} = \frac{(2{,}250)(500)(48)}{2{,}000} = \$27{,}000/\text{year}$$

$$ATTC_{DC\rightarrow PdL} = \frac{K\lambda_{DC\rightarrow PdL}}{q_{DC\rightarrow PdL}} = \frac{(2{,}250)(1{,}000)(48)}{2{,}000} = \$54{,}000/\text{year}$$

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Example 7.7.a

Annual Inventory Holding Cost (AIHC)

$$AIHC_{Brc} = \frac{hq_{DC \to Brc}}{2} = \frac{(2.5)(48)(2{,}000)}{2} = \$120{,}000/\text{year}$$

$$ATTC_{SaS} = \frac{hq_{DC \to SaS}}{2} = \frac{(2.5)(48)(2{,}000)}{2} = \$120{,}000/\text{year}$$

$$ATTC_{PdL} = \frac{hq_{DC \to PdL}}{2} = \frac{(2.5)(48)(2{,}000)}{2} = \$120{,}000/\text{year}$$

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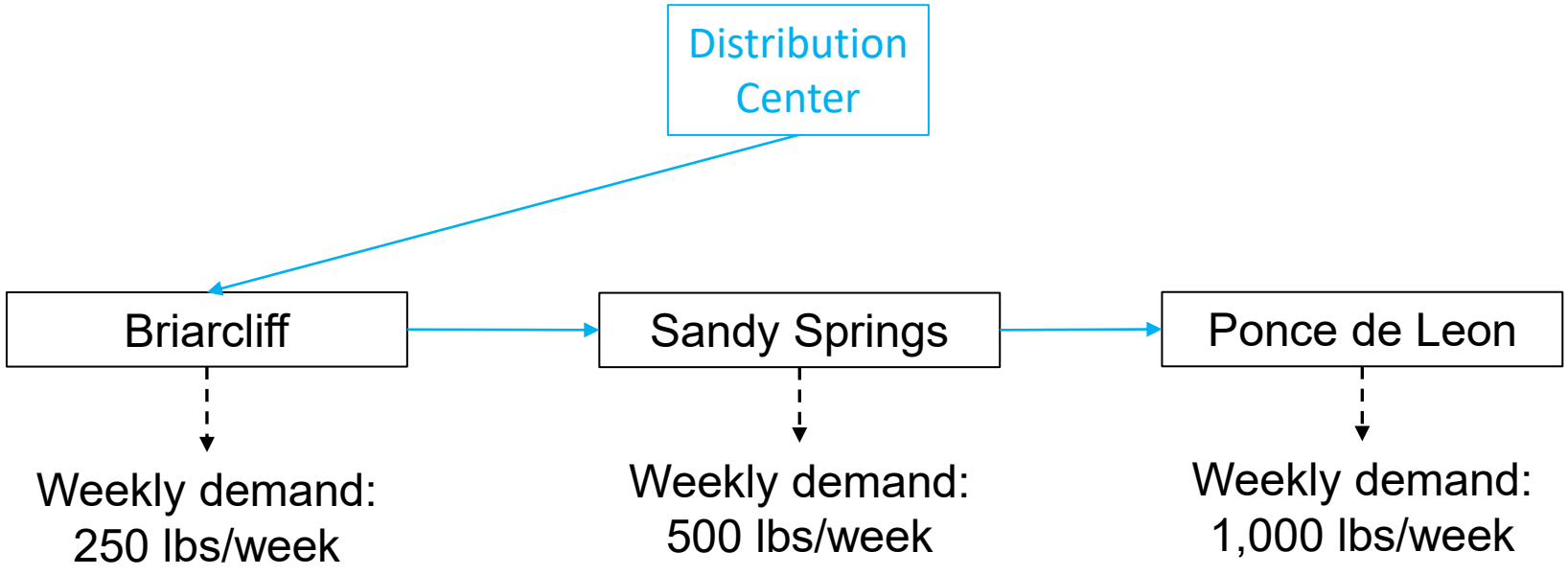
Example 7.7.b

What would the annual total logistics cost (including transportation and on-site inventory costs) be if the company were to start using trucks with a capacity of 3,500lbs and a cost of $3,500 per trip to make full truckload milk-run deliveries to stores in Atlanta?



Every link on this network has two types of costs associated with it: (1) Transportation; and (2) On-site inventory.

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Example 7.7.b

Annual Total Transportation Cost (ATTC)

$$ATTC_{DC\to Brc\to SaS\to PdL} = \frac{K\lambda_{DC\to Brc\to SaS\to PdL}}{q_{DC\to Brc\to SaS\to PdL}} = \frac{(3{,}500)(250+500+1{,}000)(48)}{3{,}500}$$
$$= \$84{,}000/\text{year}$$

Annual Inventory Holding Cost (AIHC)

$$AIHC_{DC\to Brc\to SaS\to PdL} = \frac{hq_{DC\to Brc\to SaS\to PdL}}{2} = \frac{(2.5)(48)}{2}(500+1{,}000+2{,}000)$$
$$= \$210{,}000/\text{year}$$

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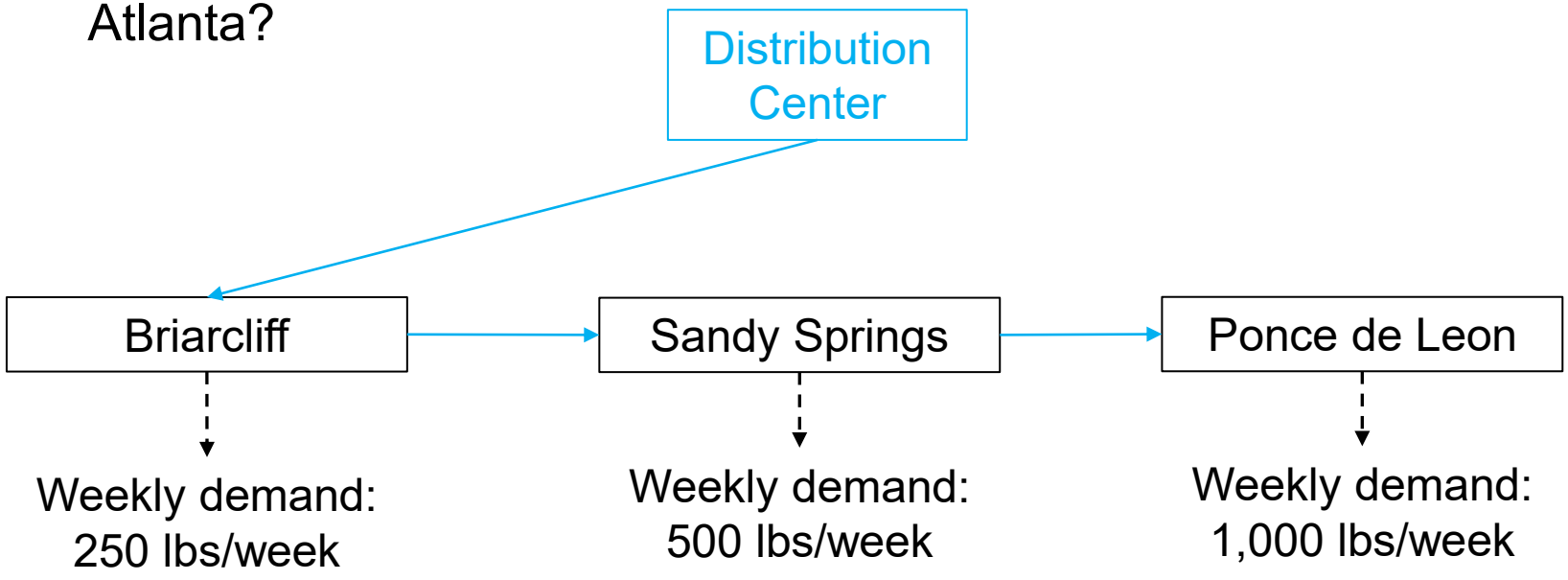
Example 7.7.c

b) What would the annual total logistics cost (including transportation and on-site inventory costs) be if the company were to start using trucks with a capacity of 3,500lbs and a cost of $3,500 per trip to make optimal shipment size for milk-run deliveries to stores in Atlanta?



Every link on this network has two types of costs associated with it: (1) Transportation; and (2) On-site inventory.

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Example 7.7.c

Optimal Shipment Size

$$q_{DC\to Brc\to SaS\to PdL} = \sqrt{\frac{2K\lambda_{DC\to Brc\to SaS\to PdL}}{h}} = \sqrt{\frac{2(3{,}500)(250+500+1{,}000)(48)}{(2.5)(48)}}$$

$$= 2{,}213.60 \text{ lbs}$$

Annual Total Transportation Cost (ATTC)

$$ATTC_{DC\to Brc\to SaS\to PdL} = \frac{K\lambda_{DC\to Brc\to SaS\to PdL}}{q_{DC\to Brc\to SaS\to PdL}} = \frac{(3{,}500)(250+500+1{,}000)(48)}{2{,}213.60}$$

$$= \$132{,}216/\text{year}$$

Annual Inventory Holding Cost (AIHC)

$$AIHC_{DC\to Brc\to SaS\to PdL} = \frac{hq_{DC\to Brc\to SaS\to PdL}}{2} = \frac{(2.5)(48)(2{,}213.60)}{2} = \$132{,}216/\text{year}$$

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Whole Foods: Comparison

	Direct shipment FTL	Milk-run FTL	Milk-run optimal
Transportation	$94,500	$84,000	$132,816
On-Site Inv.	$360,000	$210,000	$132,816
Total	$454,500	$294,000	$265,632

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Stores don’t sell your favorite product anymore. That’s on purpose

71 different versions of its Hormel Pepperoni brand.

https://www.cnn.com/2024/11/08/business/levis-dollar-general-product-choices/index.html

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Stores don’t sell your favorite product anymore. That’s on purpose

- Hormel is removing, consolidating or repackaging 25% of the items to prune unprofitable items across dozens of its brands like Spam, Applegate and Jennie-O.

- Around 80% of Hormel’s profit comes from a small number of products, like Hormel Bacon and Fire Braised-brand meats, while the rest of its tens of thousands of items often drive up costs and sit untouched in warehouses and on shelves for long periods.

- Hormel is reviewing its product lineup to invest in items with higher profit margins, improve lower-performing items and “remove production **(supply chain)** complexity”

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Stores don’t sell your favorite product anymore. That’s on purpose

Cutting down 15% clothing across its portfolio told CNN.

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Stores don’t sell your favorite product anymore. That’s on purpose

- Levi's sales through its wholesale channels - which include department stores such as Macy's and other retailers such as Walmart - fell by 19% on a constant as consumers reduced spending on clothing

- Department stores and retailers reduced their orders

- Levi’s
  - Focus on top selling items
  - Sell more clothing directly to consumers through its own stores and websites

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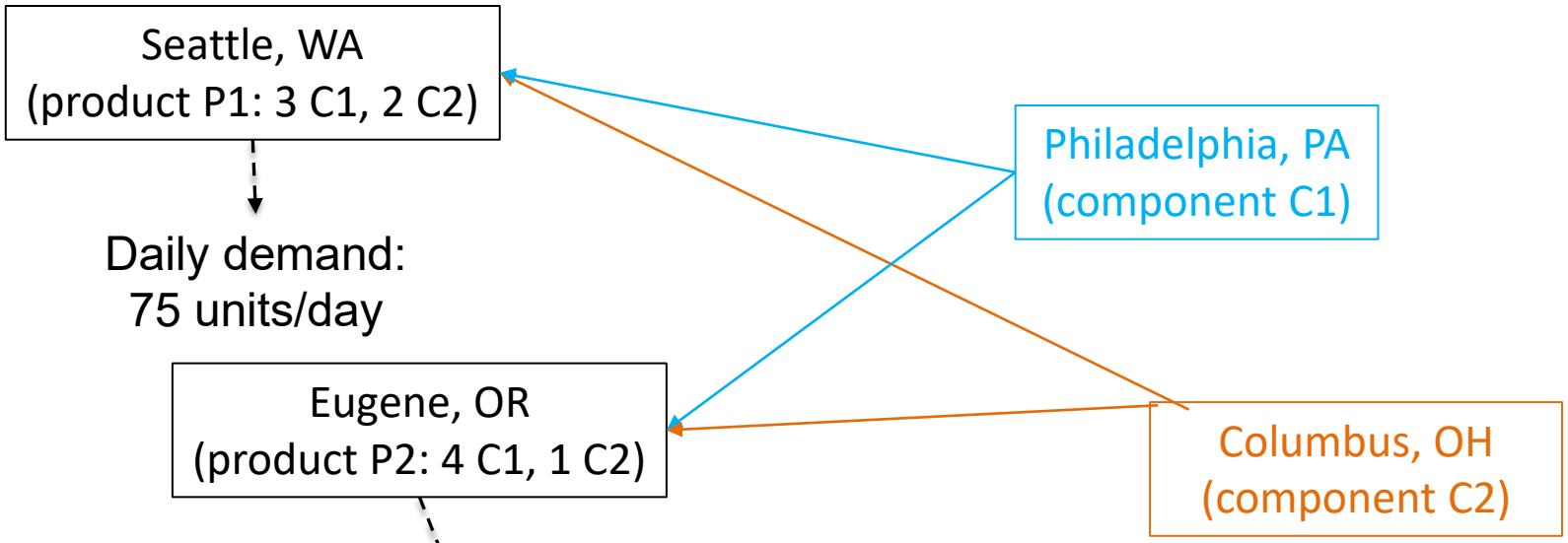
Example: 12:50 section



Every link on this network has three types of costs associated with it: (1) Transportation; (2) On-site inventory and (3) In-transit inventory.

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Example 12:50 section

- These plants operate for 360 days per year.
- The travel times from Columbus to the West Coast is 9 days, from Philadelphia to the West Coast is 10 days.
- The capacity of a truck is 5,500 units (regardless of the type of the component shipped). A shipment costs $3,000/dispatch.
- Inventory holding costs incurred while in-transit and in storage on-site are $0.10/unit/day and $0.12/unit/day, respectively.

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Example: Direct FTL

Annual Total Transportation Cost (ATTC)

$$ATTC_{Col\to Sea} = \frac{K\lambda_{Col\to Sea}}{q_{Col\to Sea}} = \frac{(3{,}000)(360)(2)(75)}{5{,}500} = \$29{,}454.55/\text{year}$$

$$ATTC_{Col\to Eug} = \frac{K\lambda_{Col\to Eug}}{q_{Col\to Eug}} = \frac{(3{,}000)(360)(1)(100)}{5{,}500} = \$19{,}636.36/\text{year}$$

$$ATTC_{Phl\to Sea} = \frac{K\lambda_{Phl\to Sea}}{q_{Phl\to Sea}} = \frac{(3{,}000)(360)(3)(75)}{5{,}500} = \$44{,}181.82/\text{year}$$

$$ATTC_{Phl\to Eug} = \frac{K\lambda_{Phl\to Eug}}{q_{Phl\to Eug}} = \frac{(3{,}000)(360)(4)(100)}{5{,}500} = \$78{,}545.45/\text{year}$$

$$\frac{(3{,}000)(360)}{5{,}500}[(2)(75) + (1)(100) + (3)(75) + (4)(100)] = \$79{,}020.45/\text{year}$$

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Example: Direct FTL

Annual Onsite Inventory Holding Cost (AOIHC)

$$AOIHC_{Col\rightarrow Sea} = \frac{hq_{Col\rightarrow Sea}}{2} = \frac{(0.12)(360)(5{,}500)}{2} = \$118{,}800/\text{year}$$

$$AOIHC_{Col\rightarrow Eug} = \frac{hq_{Col\rightarrow Eug}}{2} = \frac{(0.12)(360)(5{,}500)}{2} = \$118{,}800/\text{year}$$

$$AOIHC_{Phl\rightarrow Sea} = \frac{hq_{Phl\rightarrow Sea}}{2} = \frac{(0.12)(360)(5{,}500)}{2} = \$118{,}800/\text{year}$$

$$AOIHC_{Phl\rightarrow Eug} = \frac{hq_{Phl\rightarrow Eug}}{2} = \frac{(0.12)(360)(5{,}500)}{2} = \$118{,}800/\text{year}$$

$$(4)\left[\frac{(0.12)(360)(5{,}500)}{2}\right] = \$475{,}200/\text{year}$$

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Example: Direct FTL

Annual In−transit Inventory Holding Cost (AIIHC)

$$AIIHC_{Col\to Sea} = h\lambda_{Col\to Sea}\tau_{Col\to Sea} = (0.10)(360)(2)(75)(360)\left(\frac{9}{360}\right) = \$48{,}600/\text{year}$$

$$AIIHC_{Col\to Eug} = h\lambda_{Col\to Eug}\tau_{Col\to Eug} = (0.10)(360)(1)(100)(360)\left(\frac{9}{360}\right) = \$32{,}400/\text{year}$$

$$AIIHC_{Phl\to Sea} = h\lambda_{Phl\to Sea}\tau_{Phl\to Sea} = (0.10)(360)(3)(75)(360)\left(\frac{10}{360}\right) = \$81{,}000/\text{year}$$

$$AIIHC_{Phl\to Eug} = h\lambda_{Phl\to Eug}\tau_{Phl\to Eug} = (0.10)(360)(4)(100)(360)\left(\frac{10}{360}\right) = \$144{,}000/\text{year}$$

$$(0.10)(360)[(2)(75)(9) + (1)(100)(9) + (3)(75)(10) + (4)(100)(10)]$$
$$= (0.10)(360)(6{,}500) = \$234{,}000/\text{year}$$

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Example: Optimal LTL

Optimal Shipment Sizes

$$q_{Col\to Sea} = \sqrt{\frac{2K\lambda_{Col\to Sea}}{h_{Col\to Sea}}} = \sqrt{\frac{2(3{,}000)(2)(75)(360)}{(0.12)(360)}} = 2{,}738.61 \text{ units}$$

$$q_{Col\to Eug} = \sqrt{\frac{2K\lambda_{Col\to Eug}}{h_{Col\to Eug}}} = \sqrt{\frac{2(3{,}000)(1)(100)(360)}{(0.12)(360)}} = 2{,}236.07 \text{ units}$$

$$q_{Phl\to Sea} = \sqrt{\frac{2K\lambda_{Phl\to Sea}}{h_{Phl\to Sea}}} = \sqrt{\frac{2(3{,}000)(3)(75)(360)}{(0.12)(360)}} = 3{,}354.10 \text{ units}$$

$$q_{Phl\to Eug} = \sqrt{\frac{2K\lambda_{Phl\to Eug}}{h_{Phl\to Eug}}} = \sqrt{\frac{2(3{,}000)(4)(100)(360)}{(0.12)(360)}} = 4{,}472.14 \text{ units}$$

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Example: Optimal LTL

Annual Total Transportation Cost (ATTC)

$$ATTC_{Col\to Sea} = \frac{K\lambda_{Col\to Sea}}{q_{Col\to Sea}} = \frac{(3{,}000)(360)(2)(75)}{2{,}738.61} = \$59{,}154.10/\text{year}$$

$$ATTC_{Col\to Eug} = \frac{K\lambda_{Col\to Eug}}{q_{Col\to Eug}} = \frac{(3{,}000)(360)(1)(100)}{2{,}236.07} = \$48{,}299.02/\text{year}$$

$$ATTC_{Phl\to Sea} = \frac{K\lambda_{Phl\to Sea}}{q_{Phl\to Sea}} = \frac{(3{,}000)(360)(3)(75)}{3{,}354.10} = \$72{,}448.65/\text{year}$$

$$ATTC_{Phl\to Eug} = \frac{K\lambda_{Phl\to Eug}}{q_{Phl\to Eug}} = \frac{(3{,}000)(360)(4)(100)}{4{,}472.14} = \$96{,}598.05/\text{year}$$

$$(3{,}000)(360)\left[\frac{(2)(75)}{2{,}738.61} + \frac{(1)(100)}{2{,}236.07} + \frac{(3)(75)}{3{,}354.10} + \frac{(4)(100)}{4{,}472.14}\right] = \$276{,}499.82/\text{year}$$

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Example: Optimal LTL

Annual Onsite Inventory Holding Cost (AOIHC)

$$AOIHC_{Col\to Sea} = \frac{hq_{Col\to Sea}}{2} = \frac{(0.12)(360)(2{,}738.61)}{2} = \$59{,}153.98/\text{year}$$

$$AOIHC_{Col\to Eug} = \frac{hq_{Col\to Eug}}{2} = \frac{(0.12)(360)(2{,}236.07)}{2} = \$48{,}299.11/\text{year}$$

$$AOIHC_{Phl\to Sea} = \frac{hq_{Phl\to Sea}}{2} = \frac{(0.12)(360)(3{,}354.10)}{2} = \$72{,}448.56/\text{year}$$

$$AOIHC_{Phl\to Eug} = \frac{hq_{Phl\to Eug}}{2} = \frac{(0.12)(360)(4{,}472.14)}{2} = \$96{,}598.22/\text{year}$$

$$(0.12)(360)\left[\frac{2{,}738.61}{2} + \frac{2{,}236.07}{2} + \frac{3{,}354.10}{2} + \frac{4{,}472.14}{2}\right] = \$276{,}499.87/\text{year}$$

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Example: Optimal LTL

- Suppose a trucking company in Ohio made an offer to make milk-run pick ups from Columbus and Philadelphia and drop components in Seattle and Eugene.
- Using trucks with capacity 5,500 units of each component type
- $1 per dispatch for full truckloads
- The lead time from Columbus to Philadelphia is 1 day, from Philadelphia to Seattle is 3 days, and from Seattle to Eugene is 2 days.
- What are the total logistics costs associated with full truckload milk-run deliveries?

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Example 7.7.c

Annual Total Transportation Cost (ATTC)

$$ATTC_{Col\to Phil\to Sea\to Eug} = \frac{K\lambda_{Col\to Phil\to Sea\to Eug}}{q_{Col\to Phil\to Sea\to Eug}} = \frac{(1)[(2)(75)+(1)(100)+(3)(75)+(4)(100)]\,(360)}{5{,}500}$$

Annual Inventory Holding Cost (AIHC)

$$AIHC_{DC\to Brc\to SaS\to PdL} = \frac{hq_{Col\to Phil\to Sea\to Eug}}{2} = \frac{(0.12)(360)(5{,}500)}{2}$$

Annual In−transit Inventory Holding Cost (AIIHC)

$$AIIC_{Col\to Phil\to Sea\to Eug} =$$
$$(0.10)(360)(360)\left[(2)(75)\left(\frac{1}{360}\right)+\ (1)(100)\left(\frac{1+3+2}{360}\right)+(3)(75)\left(\frac{2}{360}\right)+(4)(100)\left(\frac{3+2}{360}\right)\right]$$

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